The next-to-leading trace coefficient of a two-peg web world

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MathDB 332639. Full all-size enumeration, with the source index corrected.

Task 1: the intended trace enumeration

Following Dukes and White [1, Section 7], let W_n be the web world on two pegs with n edges. Its diagrams are

$$D_\pi = \{(1, 2, i, \pi_i) : 1 \le i \le n\}, \qquad \pi \in S_n.$$

A k-colouring is a surjection from the edges to $[k]$. Reconstruction retains the relative endpoint orders within each colour and stacks the colour classes in increasing colour order. The coefficient $[x^k]M^{(W_n)}_{D_\pi, D_\sigma}(x)$ counts colourings of D_π whose reconstruction is D_σ. Thus the trace counts pairs consisting of a diagram and a colouring reconstructing that same diagram.

Conjecture 29 asks for the next-to-leading coefficient

$$a_n = [x^{n-1}] \operatorname{tr} M^{(W_n)}(x), \qquad n \ge 2.$$

The source's Figure 7 gives $2, 8, 42, 264, 1920, 15840$ for $n = 2, \dots, 7$ and identifies them with the sequence A121635. Its displayed formula has a one-place index shift. The theorem below proves the intended enumeration for every n, including the precise relation to that sequence.

Theorem 1. *For every integer $n \ge 2$,*

$$\boxed{a_n = (n-1)!\left(\binom{n}{2} + 1\right) = \frac{(n-1)!(n^2 - n + 2)}{2}.} \tag{1}$$

More precisely, if $\mathrm{id}_n = (1, 2, \dots, n)$ and $\tau_j = (1, \dots, j-1, j+1, j, j+2, \dots, n)$, then

$$[x^{n-1}]M^{(W_n)}_{D_\pi, D_\pi}(x) = \begin{cases} \binom{n}{2}(n-1)!, & \pi = \mathrm{id}_n, \\ (n-2)!, & \pi = \tau_j, \quad 1 \le j < n, \\ 0, & \textit{otherwise.} \end{cases} \tag{2}$$

Proof. An onto colouring of n edges with $n-1$ colours has exactly one class of size two; every other class is a singleton. A singleton reconstructs to one edge. The two-edge class reconstructs either to the permutation 12, when its endpoints have the same relative order on the two pegs, or to 21, when their relative orders are opposite.

Stacking these blocks gives either id_n or a single adjacent transposition. Hence no other permutation can be equal to its reconstruction, proving the last case of (2).

For D_{id_n}, every pair has the same endpoint order, so every surjective colouring reconstructs the identity. There are $\binom{n}{2}$ choices of the two-edge class, followed by $(n-1)!$ bijections from the resulting $n-1$ classes to the labelled colours. This is the first case of (2).

For D_{τ_j}, the unique pair with opposite endpoint orders consists of the edges whose left endpoints are $j, j+1$. They must be the two-edge class: any other choice reconstructs the identity. If that class has colour h, exactly $h-1$ singleton blocks precede it, so the reconstructed transposition is τ_h. Thus reconstruction equals τ_j exactly when the two-edge class has colour j. The other $n-2$ edges may receive the remaining $n-2$ colours in any bijective order, giving $(n-2)!$ colourings. This proves the second case of (2), including $n=2$ with $0!=1$.

There are $n-1$ adjacent transpositions. Summing (2) gives

$$a_n = \binom{n}{2}(n-1)! + (n-1)(n-2)! = (n-1)!\left(\binom{n}{2}+1\right),$$

as claimed. □

The index and the sequence identification

The known formula recorded for the polyomino sequence [2] is

$$B_m = \frac{(m-2)!(m^2-3m+4)}{2} \quad (m \geq 2), \qquad B_1 = 1.$$

Equation (1) proves $a_n = B_{n+1}$ for every $n \geq 2$. Equivalently, for every $m \geq 3$,

$$[x^{m-2}] \operatorname{tr} M^{(W_{m-1})}(x) = \frac{(m-2)!(m^2-3m+4)}{2}. \tag{3}$$

The unshifted printed formula gives 1 at $n=2$, whereas the source itself gives 2. Merely observing that discrepancy would not solve the intended counting question. The proof above instead determines all diagonal contributions and proves the source's intended sequence identification at every size. The polyomino enumeration formula is attributed to the existing sequence record; no new polyomino theorem is claimed.

This proof uses the definition of reconstruction directly. Theorem 28 of the source gives a broader reduction to words; the general enumeration in Problem 25 remains outside this result.

Tasks 2–5: proposed mathematical follow-ups

Task 2: Determine a closed formula for $[x^{n-2}] \operatorname{tr} M^{(W_n)}(x)$ for all $n \geq 3$, treating the cases of one three-edge class and two two-edge classes.

Task 3: For each fixed deficit d, express $[x^{n-d}] \operatorname{tr} M^{(W_n)}(x)$ by a finite classification of non-singleton blocks, and determine its dependence on n.

Task 4: Construct an explicit bijection from the pairs counted in (1) to deco polyominoes of height $n+1$ with no two-cell column starting at level zero, explaining the index shift.

Task 5: Determine the next trace coefficients refined by the number of inversions of the reconstructed permutation, and derive recurrences for the resulting bivariate polynomials.

References

[1] M. Dukes and C. D. White, *Web matrices: structural properties and generating combinatorial identities*, Electronic Journal of Combinatorics **23**(1) (2016), P1.45, Section 7, pp. 21–23. Final journal version; arXiv:1603.01589v1.

[2] E. Deutsch and subsequent contributors, OEIS A121635, sequence and formula for deco polyominoes; accessed 22 September 2026. The web-matrix references in this entry are the conjectural papers in [1].